

Team 7 - E-Waste Battery Extraction

CV-23 Part B: MoveIt Integration, Safety Validation & Robot Execution

Experimental Method and Results

ROS 2 Jazzy, Gazebo, MoveIt, Kinova pick-place integration and safety validation

Ubuntu 24.04 on WSL2

Robotics integration and validation report

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1 Purpose of CV-23 Part B

Part B extended the Part A ROS 2 consumer into the existing robot-planning path. The objective was to verify that a downstream robotics component could consume the CV target, solve the existing pick-place motion sequence, reject unsafe input, and complete a controlled Gazebo pick-place run.

The final tested chain was:

```
/cv_target_pose → pick-place consumer → MoveIt IK → Kinova arm → Gazebo verification
```

The work deliberately separated downstream integration correctness from the known upstream detector bounding-box problem. The live CV pose was tested through MoveIt, but robot execution was only enabled for a controlled target that matched the verified simulated battery position.

2 Existing Robotics Integration Point

A search of the supplied simulation code identified the existing script:

```
team7_sim_mark2_v1.0/sim/scripts/pick_place_battery.py
```

The file already documented the intended perception integration point. Its battery position was hard-coded as:

```
BATTERY_XY = (-0.15, 0.62)
```

and the main sequence later used:

```
bx, by = BATTERY_XY
```

The same script already contained the parts needed for real downstream validation:

- PoseStamped construction;
- MoveIt /compute_ik and /compute_fk service clients;
- joint trajectory publishing;
- a six-waypoint pick-place sequence;
- simulated suction attachment and release;
- final Gazebo model-position verification.

CV-23 therefore replaced the hard-coded source of `bx`, `by` rather than inventing a second robot planner.

3 CV-Aware Robot Consumer

The original script was preserved. Two separate CV-aware copies were created:

```
pick_place_battery_cv.py
pick_place_battery_cv_execute.py
```

Both subscribe to `/cv_target_pose`. The callback stores the most recent world-frame pose and rejects an unexpected frame identifier. The main routine waits for a target before building the normal pick-place waypoint list.

The CV target supplies battery X and Y. Existing scene geometry continues to define the flange heights for hover, contact and lift because those heights represent robot/tool clearance rather than a general 3-D grasp estimator.

4 Dry-Run MoveIt Integration

Before any trajectory commands were allowed, a dry-run version was used. It performed the complete six-waypoint IK chain but removed active `pp.go(...)` and suction calls.

The waypoints were:

Waypoint	XY source	World Z
hover	CV battery target	1.000 m
contact	CV battery target	0.855 m
lift	CV battery target	1.070 m
over-box	fixed drop box	0.700 m
release	fixed drop box	0.550 m
retreat	fixed drop box	0.700 m

Each solve was seeded from the previous solution so the 7-DOF arm stayed on a consistent IK branch.

5 Live CV Target Diagnostic

The live detector/consumer chain produced approximately:

```
x = 0.0158
y = 0.1531
z = 0.7700
```

The corresponding source bounding box covered a very large region of the 1280x960 image. This was the same upstream detector localisation issue observed during CV-17. When the live target was passed into the robot dry-run, MoveIt returned error code -31 at the first hover waypoint, corresponding to no IK solution. Nothing moved.

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=0.0158, y=0.1531, z=0.7700 m
DRY RUN ONLY: target XY will be tested with the existing pick/place MoveIt IK chain.
No trajectory commands will be published.

solving IK from CV target (seeded chain):
  IK FAILED hover (world z=1.000) code=-31

CV-23 DRY RUN RESULT: FAILED - waypoint 'hover' is unreachable.
Nothing moved.
terminate called without an active exception
Aborted (core dumped)
```

Figure 1: The real CV target reaches MoveIt, but the first waypoint is rejected as unreachable. No trajectory is executed.

This was a useful safety result. It proved that a bad perception target was not silently converted into robot motion.

6 Controlled Known-Good IK Validation

To isolate downstream integration from the faulty detector centroid, the established battery location $(-0.15, 0.62, 0.80)$ was published through the same `/cv_target_pose` interface.

The dry-run then solved all six waypoints successfully:

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6200, z=0.8000 m
DRY RUN ONLY: target XY will be tested with the existing pick/place MoveIt IK chain.
No trajectory commands will be published.

solving IK from CV target (seeded chain):
hover    target=(-0.1500, 0.6200, 1.000) jump=2.82 rad
contact  target=(-0.1500, 0.6200, 0.855) jump=0.27 rad
lift     target=(-0.1500, 0.6200, 1.070) jump=0.43 rad
over-box target=(0.6200, 0.1500, 0.700) jump=1.66 rad
release  target=(0.6200, 0.1500, 0.550) jump=0.42 rad
retreat  target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 DRY RUN RESULT: PASSED
All downstream pick/place waypoints have valid IK solutions.
The target came from /cv_target_pose.
NO ROBOT MOTION WAS EXECUTED.
terminate called without an active exception
Aborted (core dumped)
```

Figure 2: Controlled target passes the complete downstream MoveIt IK chain with no robot motion.

This result is stronger than simply running the original hard-coded script because the target enters through the new CV-23 ROS interface and is consumed by the modified downstream component.

7 Execution Safety Gate

The execution copy adds a second condition after IK succeeds. For the first controlled integration run, the target must also be within 3 cm of the known validation position:

```
expected_x = -0.15
expected_y = 0.62
tolerance = 0.03

if abs(bx - expected_x) > tolerance or \
    abs(by - expected_y) > tolerance:
    # abort before any pp.go() call
```

This prevents a reachable but unexpected target from triggering the arm during the controlled integration experiment.

Before execution, `ros2 topic info -v` confirmed that the manual controlled publisher was the only publisher on `/cv_target_pose`. This prevented the known bad live detector pose from competing with the test input.

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ ros2 topic info /cv_target_pose -v
Type: geometry_msgs/msg/PoseStamped
Publisher count: 1
Node name: _ros2cli_1750
Node namespace: /
Topic type: geometry_msgs/msg/PoseStamped
Topic type hash: RIHS01_10f3786d7d40fd2b54367835614bff85d4ad3b5dab62bf8bca0cc232d73b4cd8
Endpoint type: PUBLISHER
GID: 01.0f.94.70.d6.06.3d.db.00.00.00.00.00.00.07.03
QoS profile:
  Reliability: RELIABLE
  History (Depth): UNKNOWN
  Durability: VOLATILE
  Lifespan: Infinite
  Deadline: Infinite
  Liveliness: AUTOMATIC
  Liveliness lease duration: Infinite
Subscription count: 0
```

Figure 3: Pre-execution check showing a single publisher on /cv_target_pose.

8 Controlled Robot Execution

The controlled target first passed the complete IK chain and then the 3 cm safety gate. Only after both checks did the script call the existing trajectory and suction methods.

The terminal recorded the full sequence:

1. hover over the CV target;
2. dip to contact;
3. suction on;
4. lift;
5. traverse to the drop box;
6. lower over the box;
7. suction off;
8. retreat.

```

ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6200, z=0.8000 m
DRY RUN ONLY: target XY will be tested with the existing pick/place MoveIt IK chain.
No trajectory commands will be published.

solving IK from CV target (seeded chain):
hover      target=(-0.1500, 0.6200, 1.000) jump=2.82 rad
contact    target=(-0.1500, 0.6200, 0.855) jump=0.27 rad
lift       target=(-0.1500, 0.6200, 1.070) jump=0.43 rad
over-box   target=(0.6200, 0.1500, 0.700) jump=1.66 rad
release    target=(0.6200, 0.1500, 0.550) jump=0.42 rad
retreat    target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 IK CHECK: PASSED
All downstream pick/place waypoints have valid IK solutions.

SAFETY CHECK: PASSED
Executing controlled CV-23 pick/place sequence...
battery before: [-0.149998, 0.619999, 0.8]
  hover over CV target: 2.82 rad / 5.6s sim (<= 226s wall)
    reached in 11s
  dip to contact: 0.25 rad / 1.0s sim (<= 90s wall)
    reached in 2s
  SUCTION ON - battery sticks to pad
  lift: 0.43 rad / 1.0s sim (<= 90s wall)
    reached in 3s
  traverse to box: 1.66 rad / 3.3s sim (<= 133s wall)
    reached in 9s
  lower over box: 0.42 rad / 1.0s sim (<= 90s wall)
    reached in 4s
  SUCTION OFF - battery drops
  retreat: 0.42 rad / 1.0s sim (<= 90s wall)
    reached in 2s

battery before: [-0.149998, 0.619999, 0.8]
battery after : [0.707647, 0.112445, 0.039959]
CV-23 EXECUTION RESULT: PASSED - battery is IN THE BOX

```

Figure 4: Controlled CV-23 robot execution. All waypoints complete and the final check reports the battery inside the box.

The screenshot above was captured before a message-only cleanup removed the inherited “DRY RUN ONLY” banner from the execution copy. The banner was stale text; the run shown here did execute the trajectory and suction sequence, as confirmed by the subsequent motion logs and final Gazebo state.

The script reported:

```

CV-23 IK CHECK: PASSED
SAFETY CHECK: PASSED
...
CV-23 EXECUTION RESULT: PASSED - battery is IN THE BOX

```

9 Gazebo Ground-Truth Verification

The battery position was checked independently after execution:

Model	X (m)	Y (m)	Z (m)
battery_1	0.707647	0.112445	0.039960
drop_box	0.620000	0.150000	0.000000

The final planar offset from the drop-box centre is approximately 0.095 m, and the battery Z is approximately 0.040 m. This satisfies the existing script’s final acceptance test and is consistent with the terminal result.

```

ad@DESKTOP-OT45K60:~$ gz model -m battery_1 -p

Requesting state for world [ewaste_room]...

Model: [28]
- Name: battery_1
- Pose [ XYZ (m) ] [ RPY (rad) ]:
  [0.707647 0.112445 0.039960]
  [1.567650 -0.828826 1.759420]
ad@DESKTOP-OT45K60:~$ gz model -m drop_box -p

Requesting state for world [ewaste_room]...

Model: [40]
- Name: drop_box
- Pose [ XYZ (m) ] [ RPY (rad) ]:
  [0.620000 0.150000 0.000000]
  [0.000000 -0.000000 0.000000]

```

Figure 5: Post-execution Gazebo model poses for `battery_1` and `drop_box`.

10 Reachable-but-Unsafe Target Test

A second controlled target $(-0.15, 0.67, 0.80)$ was intentionally selected. It is only 5 cm away from the expected Y position and remained reachable by MoveIt. All six IK waypoints succeeded, which ensured the safety test was not merely another reachability failure.

The execution gate then rejected it because the target was outside the allowed 3 cm validation tolerance. The script printed `Nothing moved`.

```

ad@DESKTOP-OT45K60:/mnt/e/team7_project_bundle$ cd /mnt/e/team7_project_bundle
python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6700, z=0.8000 m
CV-23 target will first be validated with the existing pick/place MoveIt IK chain.
Robot motion will only begin after both IK and the safety check pass.

solving IK from CV target (seeded chain):
  hover    target=(-0.1500, 0.6700, 1.000) jump=2.74 rad
  contact  target=(-0.1500, 0.6700, 0.855) jump=0.26 rad
  lift     target=(-0.1500, 0.6700, 1.070) jump=0.45 rad
  over-box target=(0.6200, 0.1500, 0.700) jump=4.43 rad
  release  target=(0.6200, 0.1500, 0.550) jump=0.42 rad
  retreat  target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 IK CHECK: PASSED
All downstream pick/place waypoints have valid IK solutions.
SAFETY ABORT: received target is not within 3 cm of the controlled validation position.
Received (-0.1500, 0.6700); expected approximately (-0.1500, 0.6200).
Nothing moved.
terminate called without an active exception
Aborted (core dumped)

```

Figure 6: Negative safety test: a reachable target passes IK but is rejected before trajectory execution.

This demonstrates two independent layers of checking:

IK reachability $\neq$ permission to execute

11 Shutdown Correction

The first dry-run and safety-abort versions completed their useful work but exited with:

```
terminate called without an active exception
Aborted (core dumped)
```

The cause was the process exiting while a background `rclpy.spin()` thread was still alive. A dedicated shutdown function was added to call `rclpy.shutdown()`, join the spin thread, and destroy the node. The same safety-abort test then returned normally with no core dump.

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ cd /mnt/e/team7_project_bundle

python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6700, z=0.8000 m
CV-23 target will first be validated with the existing pick/place MoveIt IK chain.
Robot motion will only begin after both IK and the safety check pass.

solving IK from CV target (seeded chain):
  hover    target=(-0.1500, 0.6700, 1.000) jump=2.74 rad
  contact  target=(-0.1500, 0.6700, 0.855) jump=0.26 rad
  lift     target=(-0.1500, 0.6700, 1.070) jump=0.45 rad
  over-box target=(0.6200, 0.1500, 0.700) jump=2.90 rad
  release  target=(0.6200, 0.1500, 0.550) jump=0.42 rad
  retreat  target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 IK CHECK: PASSED
All downstream pick/place waypoints have valid IK solutions.
SAFETY ABORT: received target is not within 3 cm of the controlled validation position.
Received (-0.1500, 0.6700); expected approximately (-0.1500, 0.6200).
Nothing moved.
```

Figure 7: Repeated safety-abort test after adding clean ROS thread shutdown.

12 Interpretation and Limitation

CV-23 verifies the downstream robotics integration, but the controlled robot execution should not be misreported as a successful live YOLO-to-grasp experiment. The real detector target was forwarded correctly through the CV-23 consumer, but its oversized upstream battery bounding box produced a world position far from the actual battery and failed the robot IK check.

The controlled execution therefore served a specific engineering purpose: it isolated and verified the downstream path from ROS pose input through MoveIt, trajectory control, suction and Gazebo state validation. Correcting the upstream detector localisation remains separate work.

13 Part B Outcome

Part B demonstrated that the CV-23 target interface can be consumed by the existing planning and robot-control code. The implementation:

- replaced the hard-coded battery XY source with a ROS pose subscription;
- reused the existing MoveIt IK and trajectory logic;

- validated the full six-waypoint chain before motion;
- safely rejected an unreachable live detector target;
- safely rejected a reachable target outside the controlled tolerance;
- executed a successful controlled Gazebo pick-place sequence;
- independently verified the final battery and drop-box poses;
- exits cleanly after both success and abort paths.

A Core Commands

A.1 Dry-run validation

```
python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv.py
```

A.2 Controlled execution

```
python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
```

A.3 Post-execution Gazebo state

```
gz model -m battery_1 -p  
gz model -m drop_box -p
```